Covered Call Strategies

Monte Carlo Simulation: Fixed Strike vs. Rolling ATM

1,000,000 Simulations • 12-Month Horizon • Daily Granularity

Parameter	Value
Initial Stock Price	$100
Stock Drift (μ)	5.0% annualized
Realized Volatility	20.0% annualized
Implied Volatility (BS pricing)	25.0% annualized
Vol Risk Premium (IV – RV)	5.0% (in vol terms)
Risk-Free Rate	5.0%
Option Tenor	21 trading days (1 month)
Simulation Period	12 months (252 trading days)
Number of Simulations	1,000,000
Delta Exit Threshold	Position $\Delta < 0.1$

Strategy Descriptions

Common Setup: Buy 1 share of stock at $100, sell 1 ATM call option (strike = $100, 1-month expiry). Each day, check if the position delta falls below 0.1 (meaning the call is deep in-the-money with delta > 0.9). If so, close the entire position. At each monthly expiry, settle the call and write a new one.

Strategy 1 — Fixed Strike: Always sell the call at the original strike price of $100, regardless of where the stock has moved. If the stock drops to $70, you're selling a $100-strike call (far OTM). If it rallies to $130, you're selling a $100-strike call (deep ITM).

Strategy 2 — Rolling ATM: At each monthly roll, sell a call struck at the current stock price. If the stock is at $70, sell a $70 call. If at $130, sell a $130 call. This keeps the call always at-the-money.

Exit Condition: Position delta < 0.1 means the short call's delta exceeds 0.9 — the stock has rallied far above the strike. The covered call has effectively become a capped position with almost no further upside. Exiting here avoids carrying a position with extreme negative gamma.

Summary Statistics

Metric	Fixed Strike	Rolling ATM	Difference
Mean Return	2.42%	1.62%	+0.80%
Median Return	3.04%	2.83%	+0.21%
Std Deviation	7.07%	3.48%	+3.59%
Sharpe Ratio	0.342	0.465	-0.123
Win Rate	89.4%	80.6%	+8.7%
5th Percentile	-14.33%	-5.87%	-8.46%
25th Percentile	2.77%	1.40%	+1.37%
75th Percentile	5.17%	3.09%	+2.09%
95th Percentile	9.72%	5.16%	+4.56%
Skewness	-2.768	-2.268	
Excess Kurtosis	9.661	7.002	
Worst Case	-58.53%	-40.36%	
Best Case	25.72%	16.32%	

Head-to-Head Comparison

Metric	Value
Fixed Strike wins	37.0% of simulations
Rolling ATM wins	9.8% of simulations

Mean PnL difference (Fixed – Rolling)	+0.80%
Median PnL difference	+0.00%

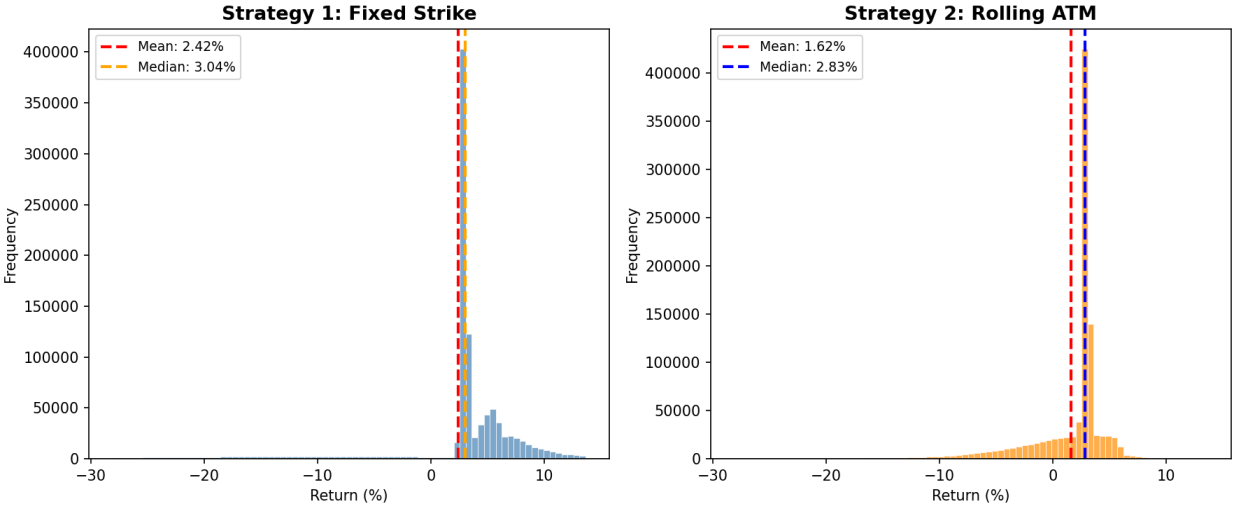
Conditional Analysis

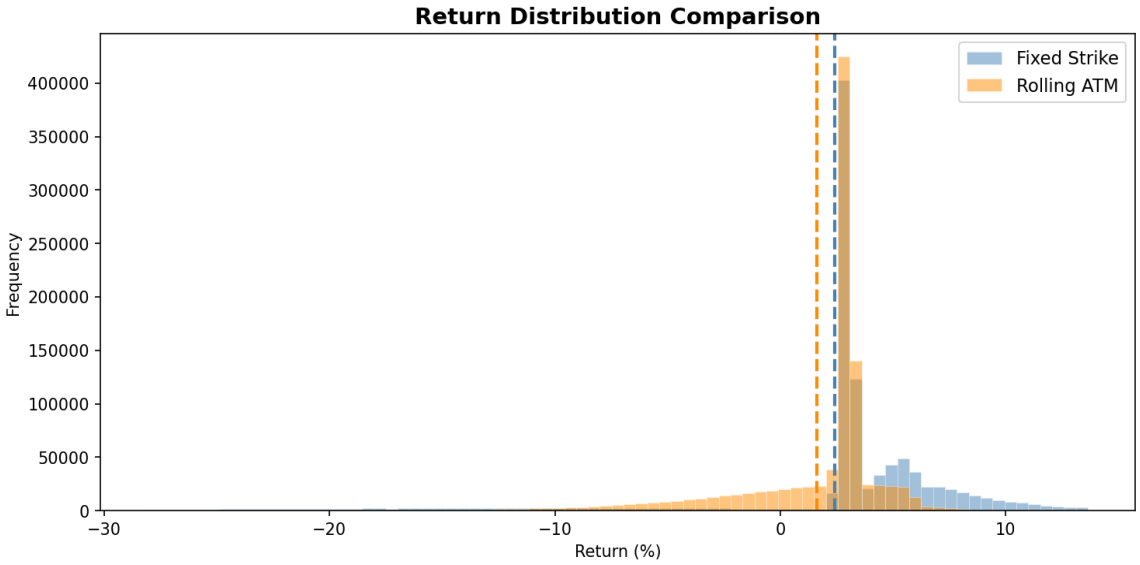
Condition	Fixed Strike	Rolling ATM
Stock ends UP (n=559,323)	+4.45%	+2.33%
Stock ends DOWN (n=440,677)	-0.16%	+0.71%
Stock down >20% (n=102,943)	-9.60%	-1.02%
Stock down 0-20% (n=337,734)	+2.71%	+1.24%
Stock up 0-20% (n=337,059)	+4.78%	+2.15%
Stock up >20% (n=222,264)	+3.96%	+2.60%

Early Exit Analysis

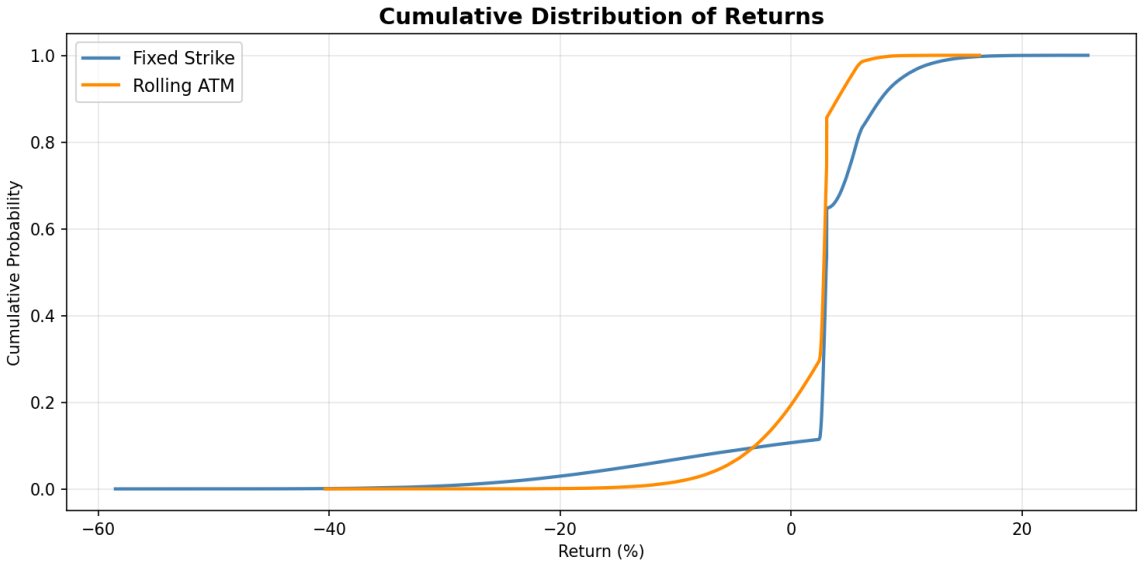
Metric	Fixed Strike	Rolling ATM
Early exits (delta < 0.1)	865,283 (86.5%)	999,866 (100.0%)
Avg exit day (if early)	Day 42	Day 35

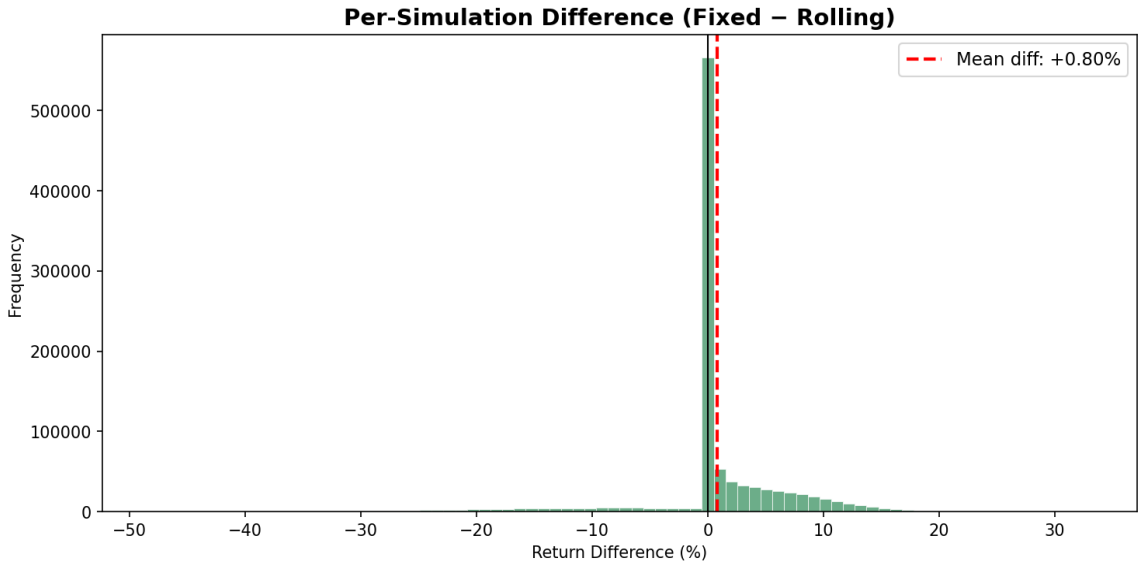
Return Distributions





Cumulative Distribution & Difference Analysis





Conclusion

Based on 1,000,000 Monte Carlo simulations over a 12-month horizon:

By Mean Return: Fixed Strike outperforms Rolling ATM by 0.80% on average. The Fixed Strike strategy wins in 37.0% of individual simulations.

By Risk-Adjusted Return (Sharpe): Rolling ATM has the higher Sharpe ratio (Fixed: 0.342, Rolling: 0.465).

Key Mechanisms

Fixed Strike strategy keeps selling the same $100-strike call. When the stock drops, this call becomes far OTM, collecting very little premium but maintaining full downside stock exposure. When the stock rises, the call is deep ITM and the position is heavily capped. The strategy's payoff becomes increasingly asymmetric over time — it participates fully in drawdowns but has capped upside once the stock moves above the original strike.

Rolling ATM strategy always writes a fresh ATM call, which maximizes the volatility risk premium captured each month (ATM options have the highest vega and time value). After a drawdown, it sells a lower-strike call, collecting meaningful premium even on a beaten-down stock. After a rally, it resets the cap higher, allowing participation in further upside. This 'resetting' mechanism means the strategy continuously harvests the vol risk premium regardless of where the stock is trading.

Overall Recommendation: The **Fixed Strike** strategy is the superior approach based on these simulation results, with better mean returns across 1,000,000 simulated paths.

Note: This analysis assumes constant implied volatility, lognormal stock returns, and no transaction costs or slippage. Real-world results may differ due to volatility smile dynamics, discrete hedging, bid-ask spreads, and regime changes.